

Seller

Attributes—

- s_id - int - unique seller id (acts as the primary key for seller entity)
- sname - varchar - seller name
- contact - varchar - unique- phone no for contact (with less than or equal to 15 digits)
- Mail_id - varchar - email address for online communication
- address - varchar - Address of the seller's shop
- add_pincode - int - Pincode of the address of the shop (6 digit)
- s_rating - int - Rating of the shop from 1 to 5

- Assumptions**
- A seller can sell multiple products with their unique pid's.
 - A seller has only one contact and mail address.
 - Have only one branch (Different branches of the same seller are considered different)

Relationships associated with the seller

Many-to-many relationship with product named as ordered
with one attribute –

Quantity - int - no of products the seller has in stock

Customer

Attributes—

- u_id - int - unique customer id (acts as the primary key for customer entity)
- first_name - varchar - the first name of the customer
- last_name - varchar - the last name of the customer
- contact - varchar - unique- phone no for contact (with less than or equal to 15 digits)

mail_id - varchar - email address for online communication
address - varchar - Address of the seller's customer
add_pincode - int - Pincode of the address of the customer (6 digits)
dob - date - date of birth of customer in format dd/mm/yyyy
gender - varchar - sex of the customer (male or female)

Assumptions -

- Have one contact and mail address
- Gender can be male or female only
- add_pincode is unique
- customer entity is identifying entity for the cart
- customer's uid is used as a foreign key for the cart

Relationships associated with the customer entity

one-to-many relationship with orders named Payment

with attributes –

p_date - date - date of payment

Mode - varchar - transaction mode whether it is upi,neft,etc

Amount - int - amount of payment

P_time - time - time of payment

other - varchar - additional information regarding payment

one-to-many relationship with Employee named Delivery

with attributes –

Delivery_time - time - time when the product is delivered to the customer

Status - varchar - status of the delivery whether it is done or

Address not found or returned or customer out of place
etc

One to one relationship with cart named cart (a weak entity)

without any specific attributes

Many-to-many relationship with product named Add_to_cart

without any specific attribute.

Product

Attributes –

pid - int - unique product id (acts as primary key for this entity)

pname - int - name of the product being sold

MRP - int - Maximum retail price of the product

Psepcification - varchar - Specification of the product

p_rating - int - product rating as given by the customers who bought
It (aggregate of the all rating)

Assumptions –

- All product of the same type has the same pid
- pname is also unique but it is varchar, so the comparison would be lengthy (hence not used as primary key)
- A product type can be sold by multiple sellers.

Relationships associated with the product entity

- Many to many relationships between product and seller. (A product can be sold by multiple sellers).
- Many to one relationship between product and category.

Category

Attributes –

-cid - category id (primary key)

-cname

Assumptions –

- A category may have multiple products.

Relationships associated with the category entity

- .One to many relationship with product.

Warehouse

Only one warehouse in a pin code. So, the pin code is used to uniquely identify a warehouse.

Relationships associated with the warehouse entity

- Many to many relationship between warehouse and seller.
- one to many relationships between warehouse and employee.

Employee

Attributes –

- eid - employee id (primary key)
- first_name (cannot be null)
- last_name
- contact - varchar - unique- phone no for contact (with less than or equal to 15 digits))
- main_id (single valued)
- address
- add_pincode (pincode of warehouse where employee works)
- dob
- designation
- gender

Assumptions –

- All employee works at only one warehouse.
- only one contact no. and mail id of each employee is available.

Relationships associated with the employee entity

- Many to many relationship between employee and customer. (A employee can deliver a order to multiple customers and a customer can get their order delivered form multiple employees).
- Many to one relationship between employee and warehouse (Each employee can work at only one warehouse).

Cart (weak entity)

Attributes –

- pid - id of products which are contained in the cart
- add_pincode (pincode of warehouse where employee works)
- sid - id of sellers whose products are in cart
- nos- quantity of a product

Assumptions –

- Cart do not have independent existence . Customer is it's identifying entity.
- A cart belongs to only one customer.

Relationships associated with the employee entity

- one to one relationship between cart and employee.
- Customer acts as identifying entity for cart.

Orders

Attributes –

- oid - order id (primary key)

- o_time - time of order
- amount - Amount to be paid by the customer.

Assumptions –

- A order belongs to only one customer.
- A order may have multiple products.

Relationships associated with the Orders entity

- Many to one relationship with customer (A customer can have more than one order).